

LIDAR manual for Operators

Version 1.0

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Document History

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Table of Contents

Document History.....	1
Co-registration.....	1

Co-registration

If your project will involve source-space analysis, you will need to collect two LIDAR scans to co-register with the participant's structural MRI scan. The first scan is collected in the MEG control room, and should include the entirety of the participants head. The second is collected once the participant is positioned in the MEG chair, and the helmet is in place with the sensors inserted. The second LIDAR scan should usually be collected before starting to record MEG data, but could also be recorded after the MEG data recording.

Get your participant comfortably seated in the MEG control room and start up the LIDAR laptop.

Power up the LIDAR camera by plugging the USB cable from the camera into the battery that is built into the camera handle. The indicator LED on the back of the camera should briefly turn blue, before flashing red until the camera is fully powered up, which is indicated by the LED turning green.



Figure 1: LIDAR camera, the battery is integrated in the handle and the 'output' port in the handle should be connected to the camera to power up the device

The LIDAR should automatically connect to the laptop via WIFI, if it does not, click on the globe icon in the bottom right hand corner of the desktop and select the POP3-REVO from the list of available WIFI connections, and click connect.

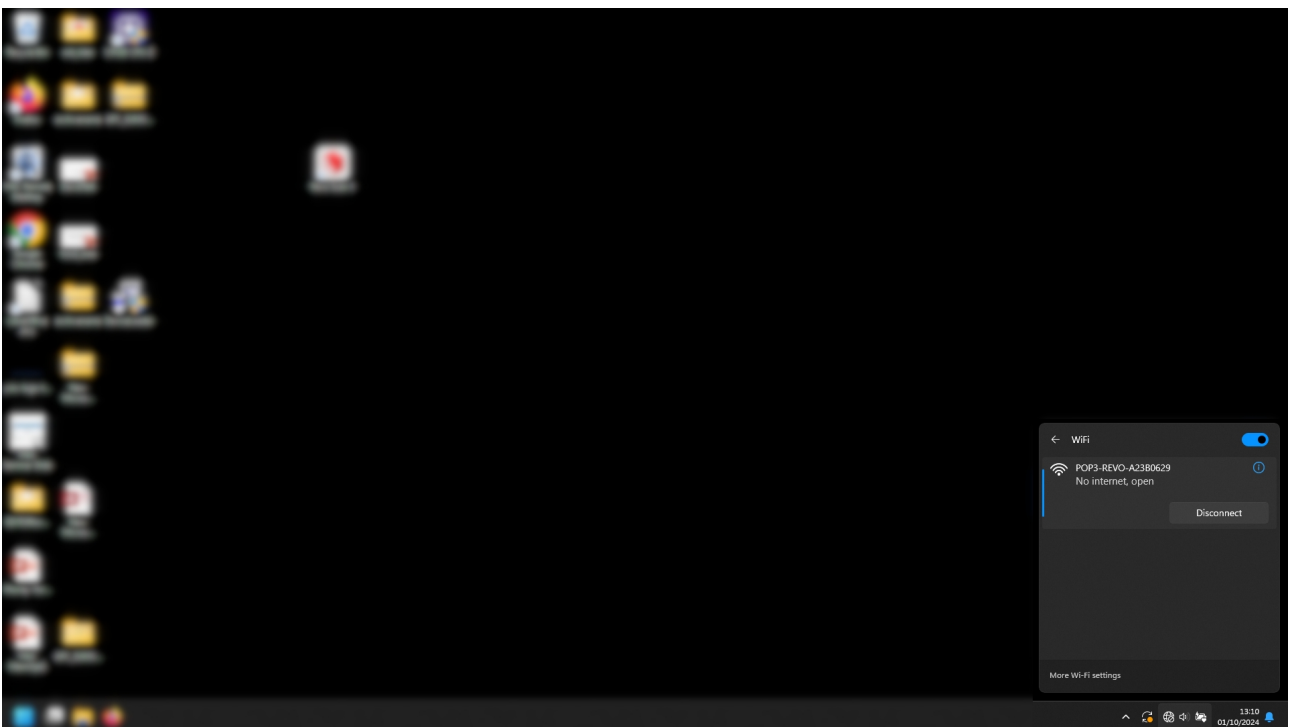


Figure 2: LIDAR laptop showing with wifi connection to camera device

Open the LIDAR software by double-clicking on the 'Revo Scan 5' icon on the desktop.



Figure 3: Desktop of LIDAR control PC

Wait for the LIDAR camera to connect to the software, when it has, you will see a green tick symbol as shown below:

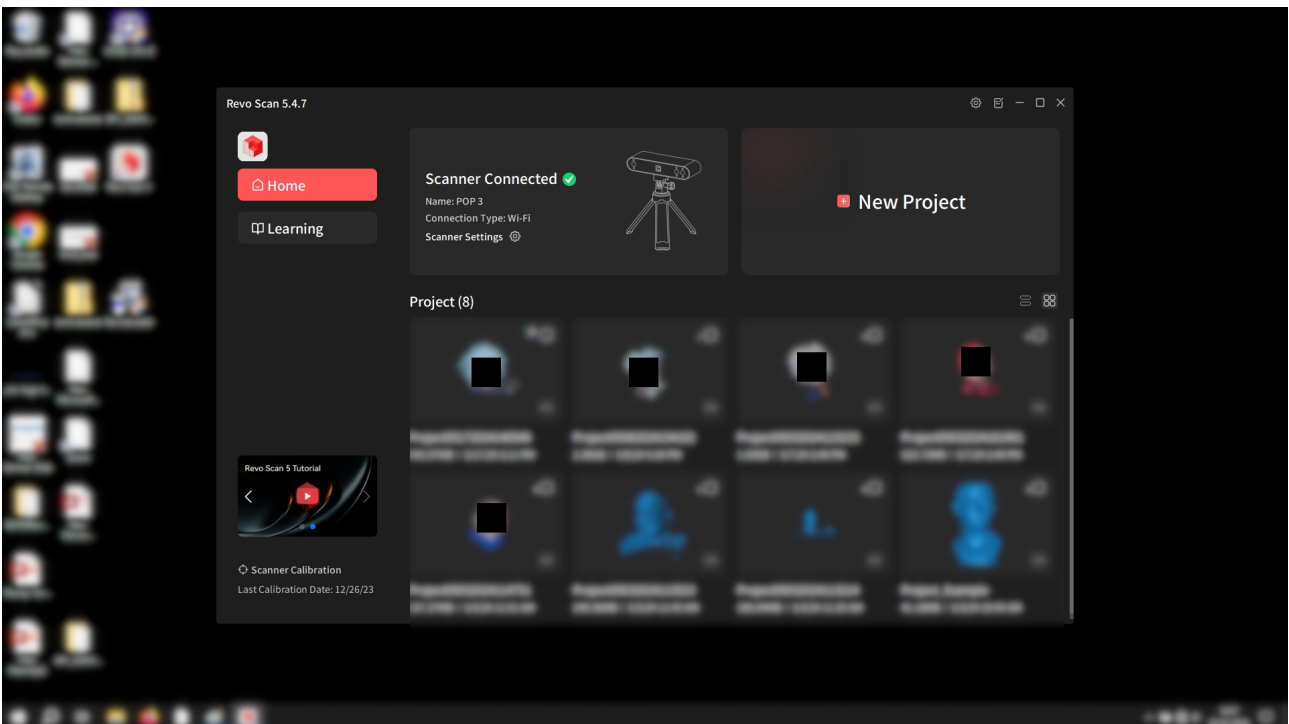


Figure 4: LIDAR camera successfully connected to Revo Scan software

Click 'New Project', and a dialogue will appear for you to enter a project filename, leave the auto-

filled value which will be 'Project' followed by a timestamp, and click the orange 'New' button.

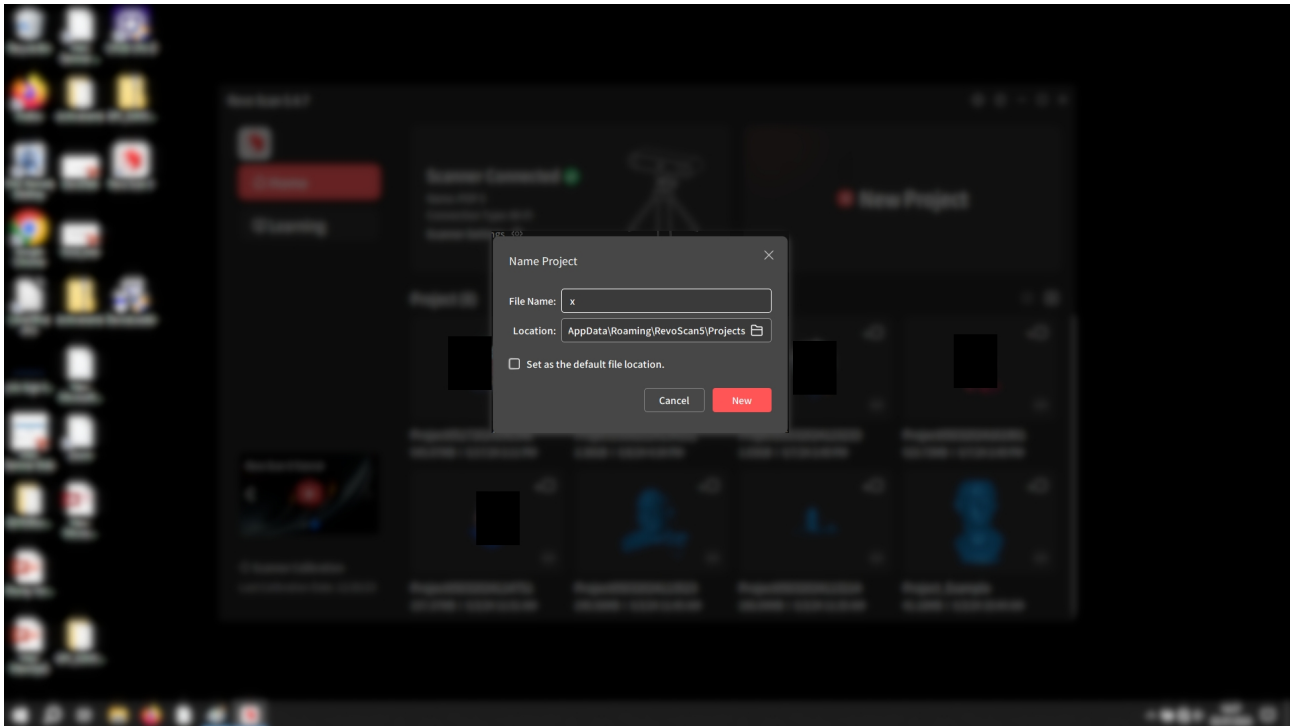


Figure 5: Naming the LIDAR acquisition project

The main LIDAR interface will now open. In the toolbar on the right hand side of the screen, click on the 'Object Type' menu, and select 'face'.

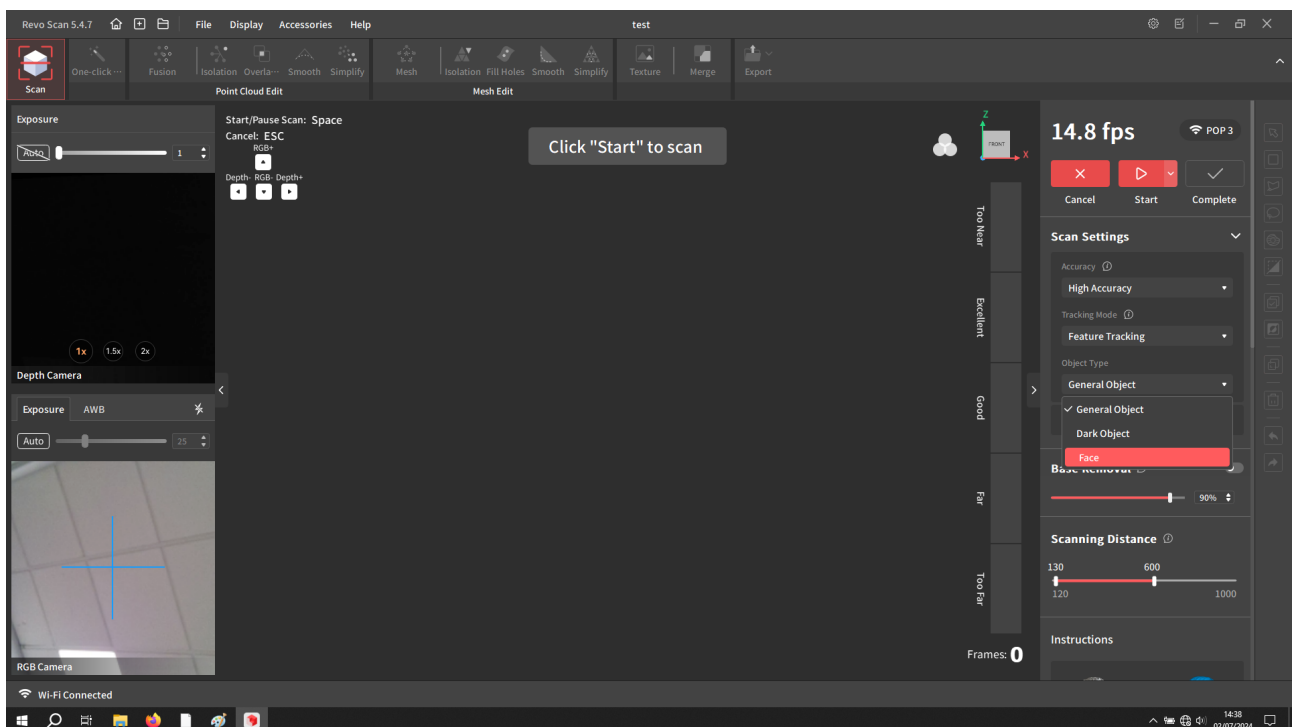


Figure 6: Select 'Object Type' as Face

Ensure that the camera is pointing away from anyone's face (eg point it at a wall) and check that no-one in the room is photosensitive, then switch on the 'Color Scanning' toggle in the toolbar to the right of the screen. The camera will generate a bright flickering light.

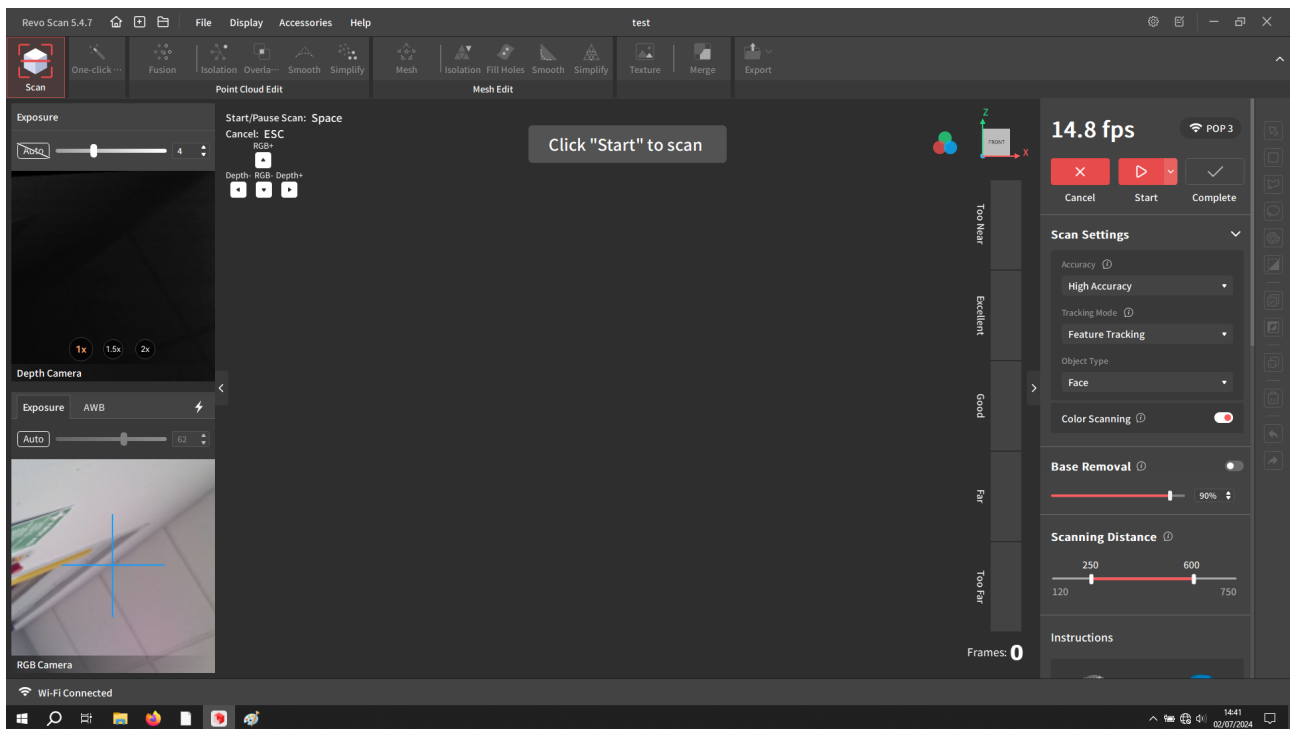


Figure 7: 'Color Scanning' switched on - this will cause a bright flickering from the camera

Ask the participant to close their eyes, then aim the LIDAR camera at their right ear at a distance of about 35 cm. A histogram should appear to the right of the screen indicating whether you are at a good range for scanning, adjust your distance so that histogram is within the Excellent – Good range, the histogram should be green in this range.

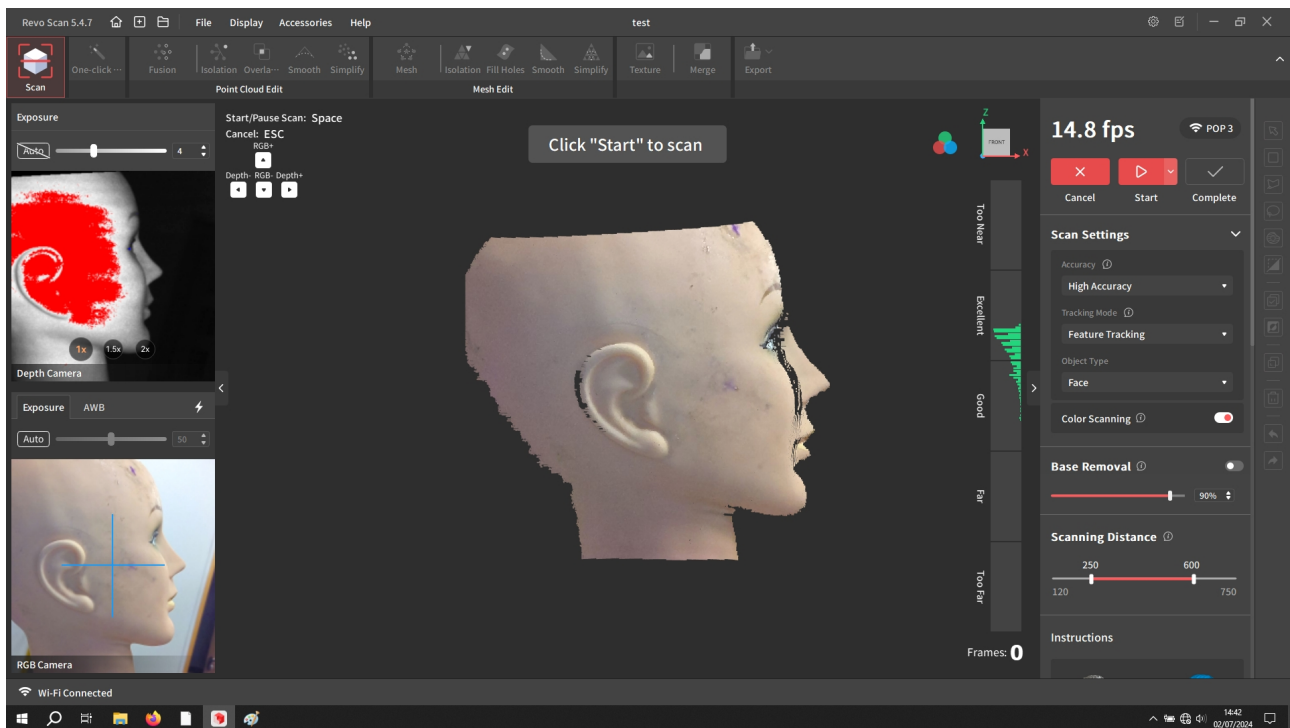


Figure 8: LIDAR interface indicating a good starting point for acquisition – the crosshair is pointing to the right of the participant's head, and the range-finding histogram is green

With the histogram green, and the crosshair in the lower left corner of the screen pointing to the participant's right ear, you have found a good starting point for the acquisition. Press the triangle shaped icon at the rear of the LIDAR camera to start collecting a scan.



Figure 9: LIDAR controls, press the triangle symbol in the middle of the device to start and pause scanning.

Slowly move the camera around the participant's head, moving to across their face and finishing at their left ear. Monitor the range-finding histogram to ensure it stays green, a 3D image should start to build up in the middle of the screen. Try to fill any gaps in the 3D scan by pointing the LIDAR camera at any missing areas. The scan may look relatively poor in quality at this stage. The important thing is that the camera has moved smoothly and remained in range so that there are no discontinuities in the scan. If a major distortion occurs, press the triangle button to pause the scan, click the orange cancel button in the right-hand tool bar, and then click 'New Scan' to start another acquisition.

Once you have a complete 3D scan of the participant's head, press the triangle button on the camera, and then click the orange 'Complete' button in the right-hand toolbar of the control PC.

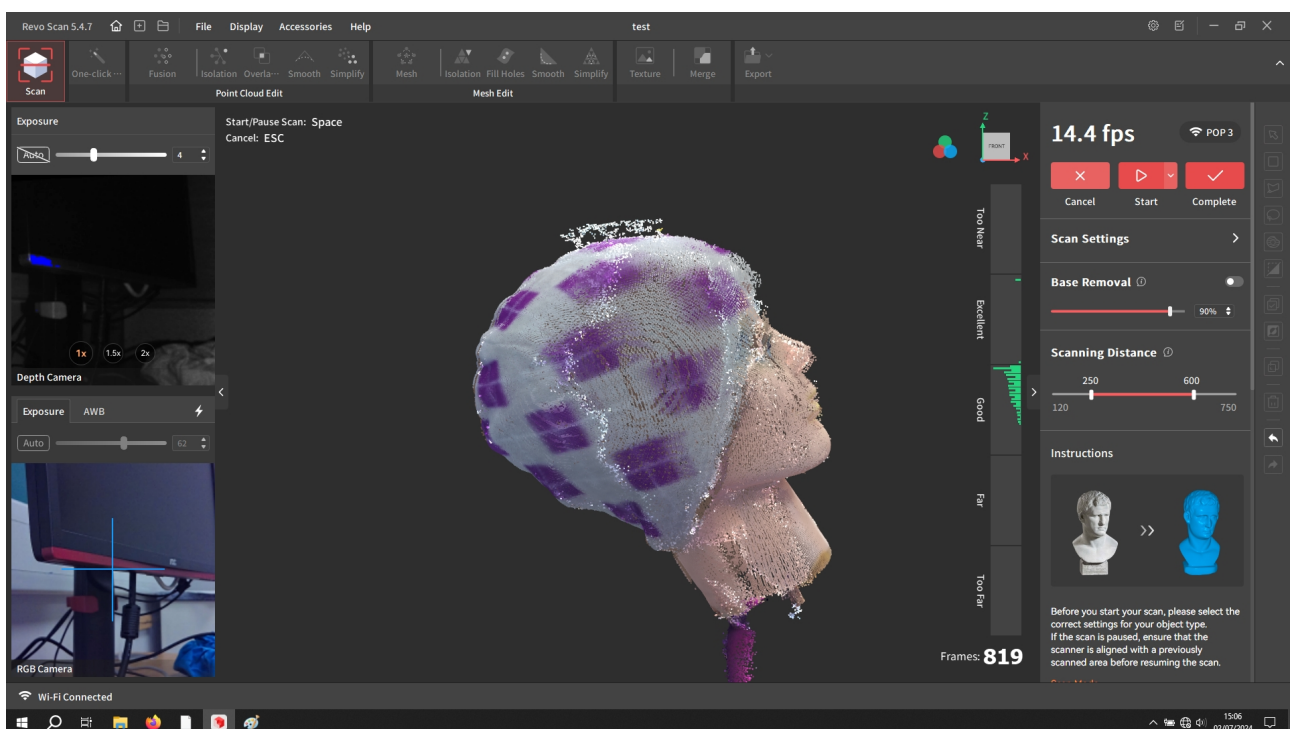


Figure 10: Completed 3D scan - the scan may look poor at this stage, but should be fine as long as the head is completely scanned with no large distortions.

Next, click on the 'One click...' icon towards the top left of the LIDAR software to move to a processing tab like the one shown below:

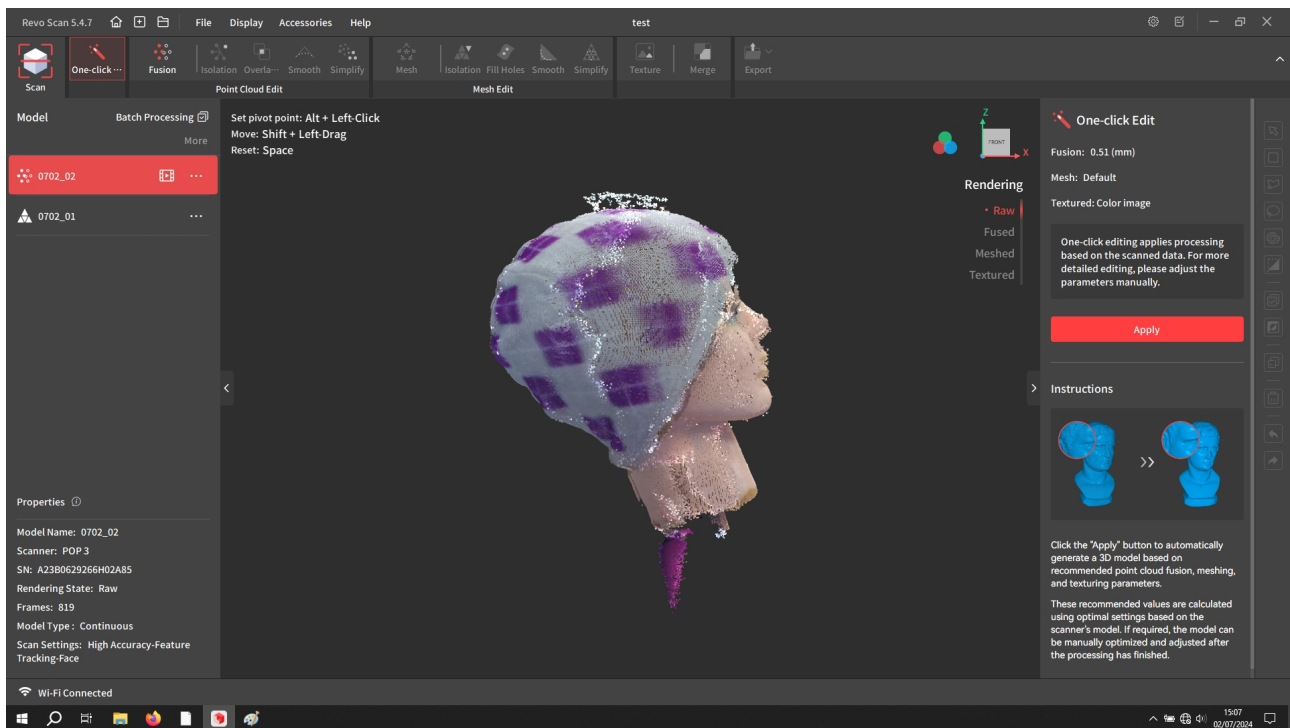


Figure 11: One-click processing tab

Click on the orange 'apply' button in the right hand toolbar. The software will start processing the 3D scan and you should see a timing dialogue. This should take 5 minutes or less, but you can continue preparing the participant for the MEG recording now.

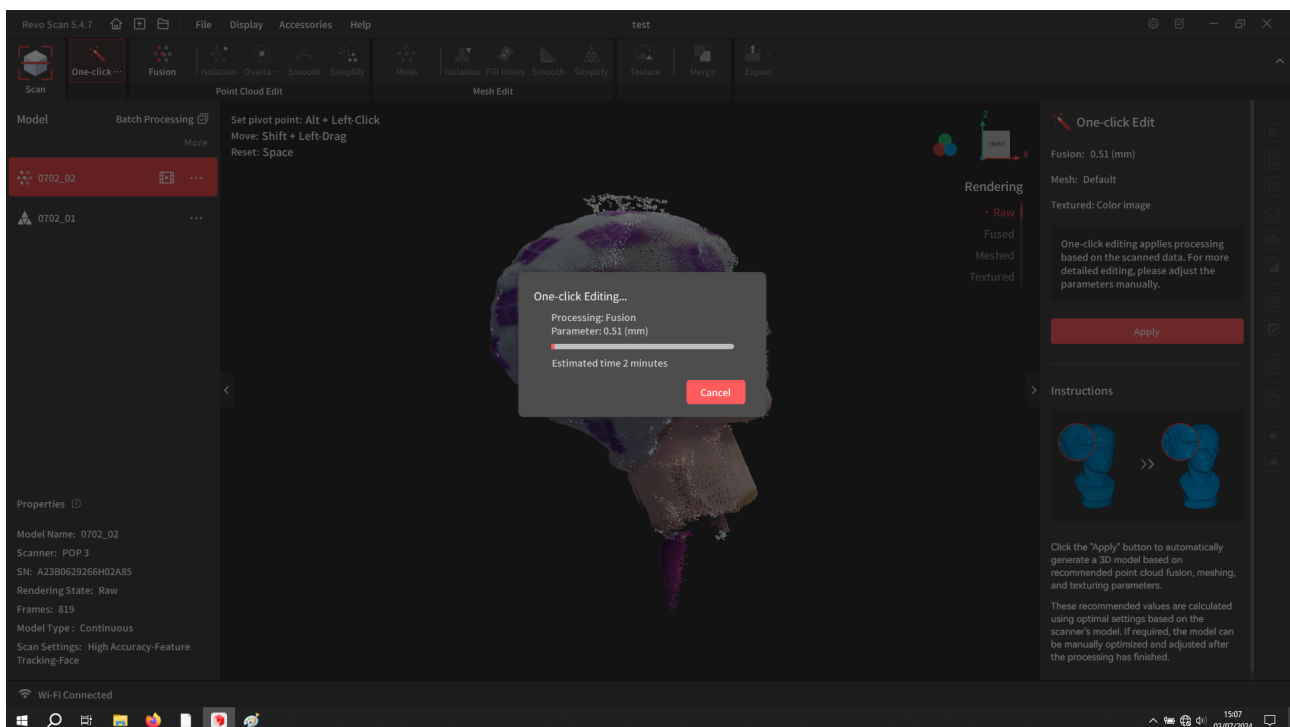


Figure 12: LIDAR software performing processing

When the processing is complete, you should see a clear 3D scan of the participant in the main interface window. Click on the 'Export' icon towards the top of the screen and select 'Mesh Model'. In the exporting dialogue, leave the auto-filled filename, but add a unique identifier (eg subject number) for the participants to the start of the filename, then export the data to the desktop.

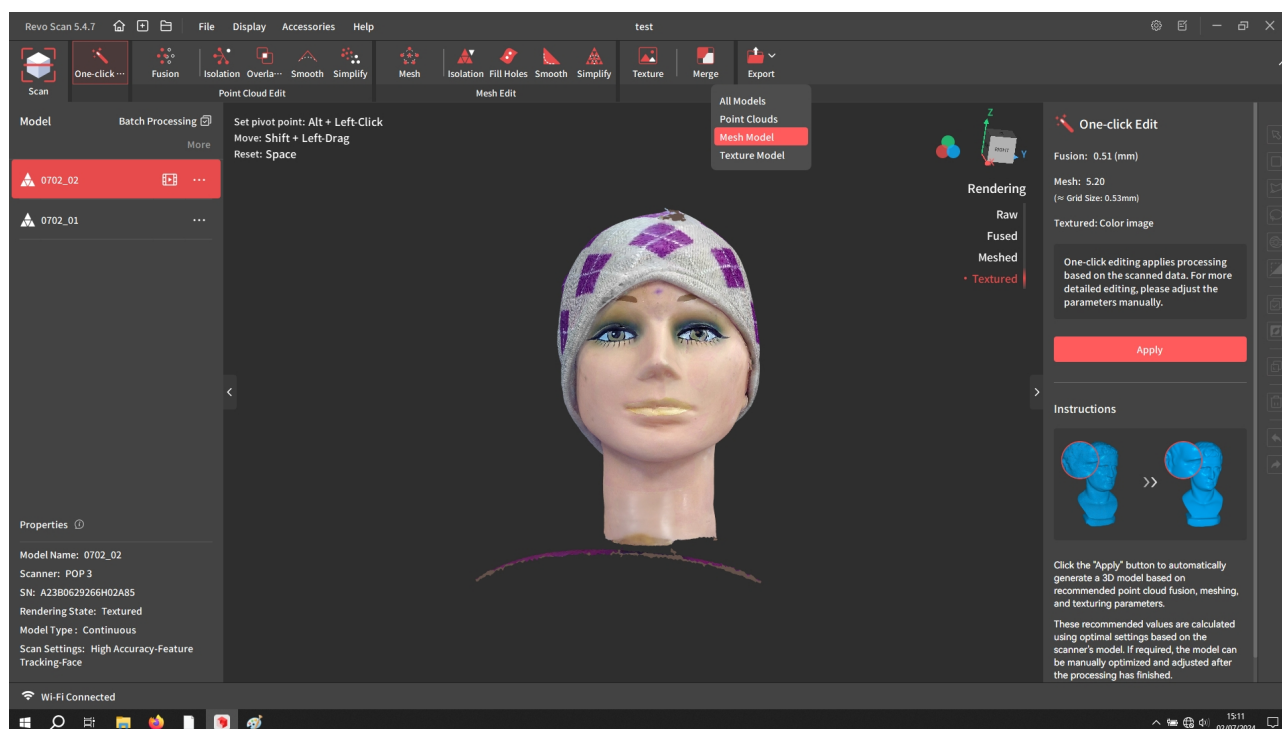


Figure 13: Exporting the finished scan

When you have finished, close the Revo Scan software by clicking on the cross in the top-right corner of the screen.

Ensure all LIDAR equipment is removed from the magnetically shielded room prior to acquiring MEG data – the magnetic fields generated by this equipment are far larger, and will exclude, the neuronal sources that you wish to measure.